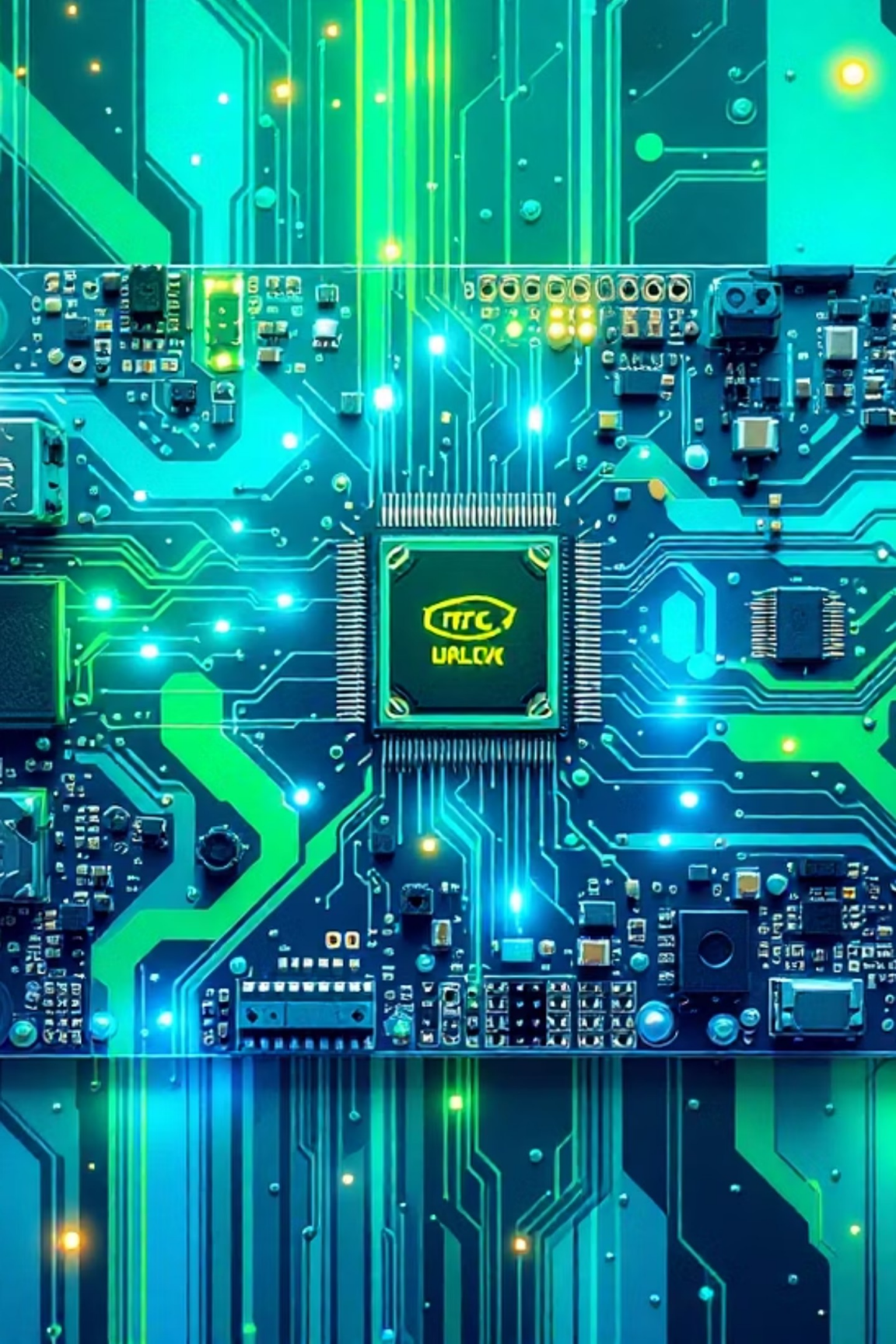




Zephyr Workshop – Kconfig + DeviceTree

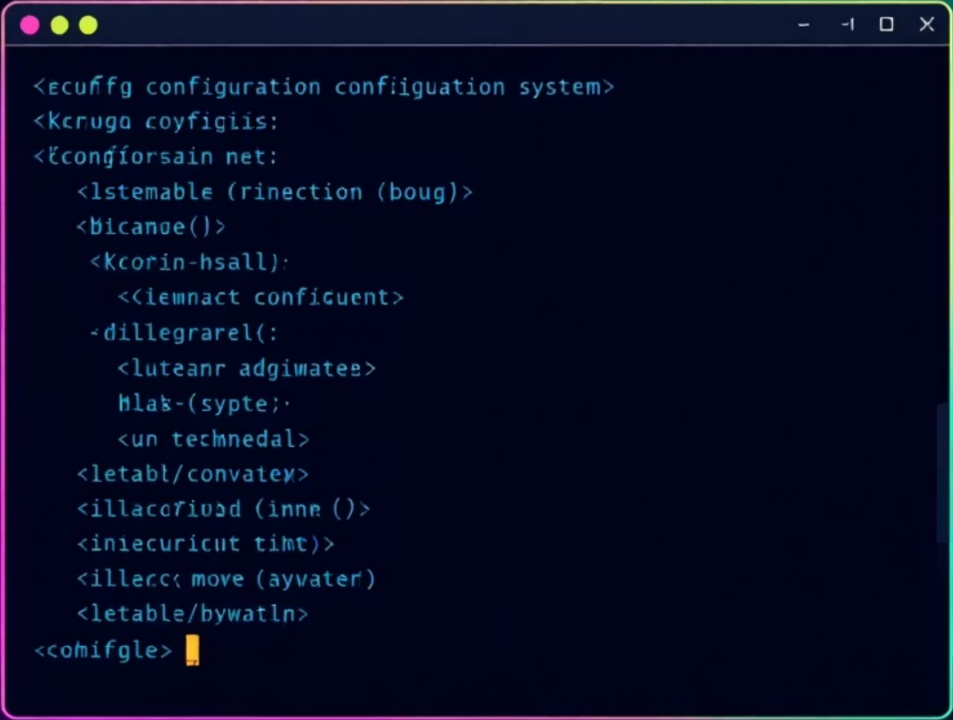
Understanding how Zephyr RTOS configures features and hardware

What is Kconfig?

Configuration system for:

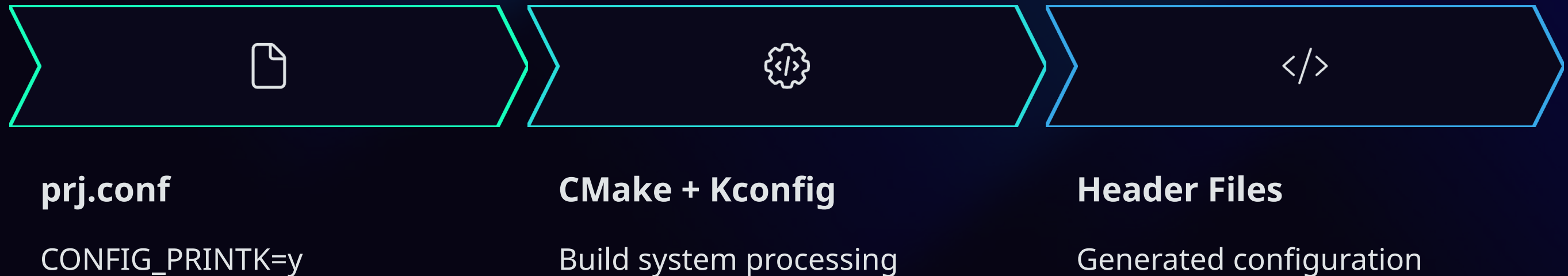
- System features
- Drivers
- RTOS options

Options start with 'CONFIG_'



```
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  <lstemable (rinection (boug)>
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  <illaco7i03d (inne ())>
  <iniecuricut timt)>
  <illercc move (ayvater)>
  <letable/bywatln>
<comifgle>
```

Kconfig – How it Works

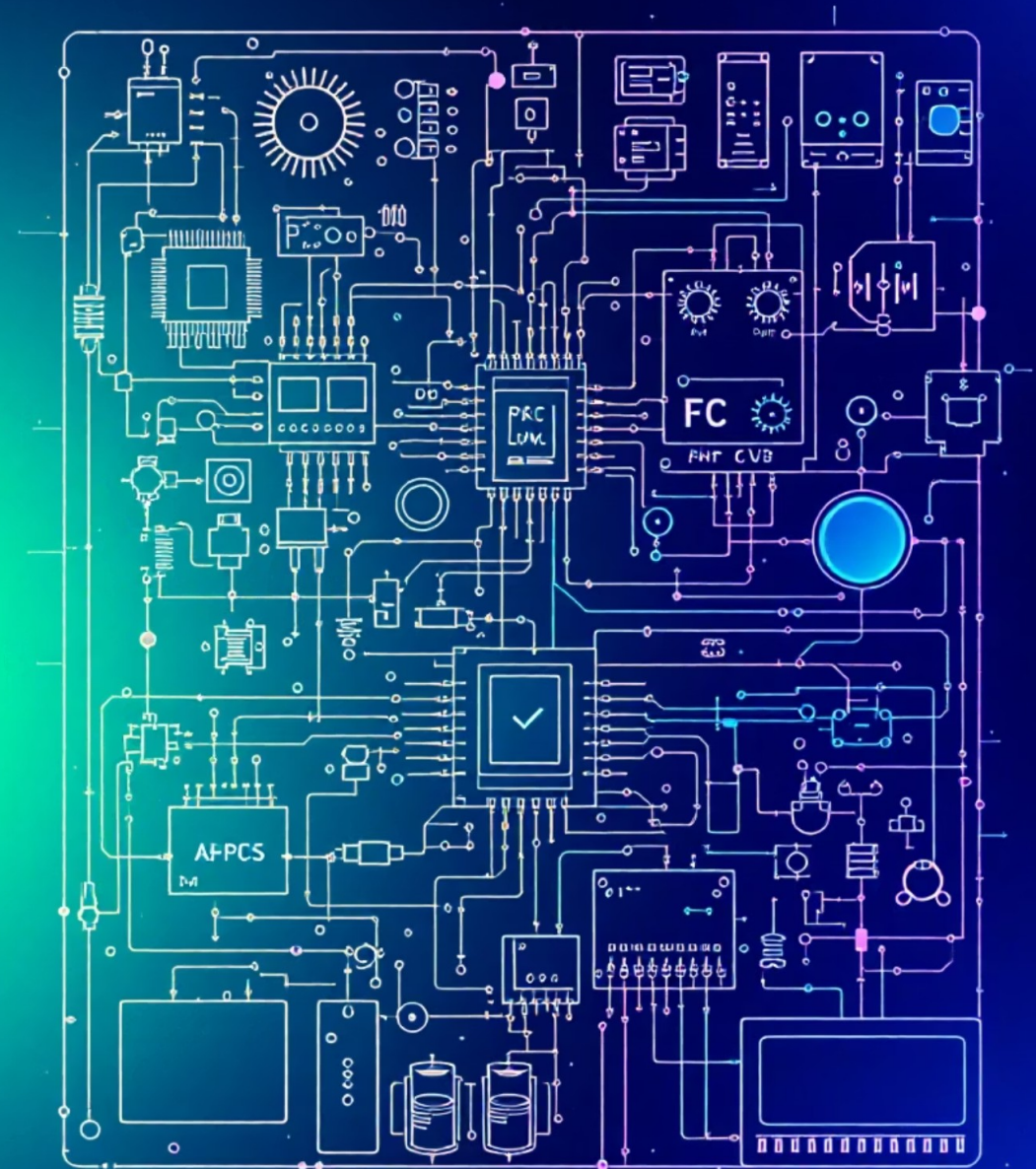


Use 'west build -t guiconfig' to explore

What is DeviceTree?

Hardware description language that:

- Describes hardware layout and connections
- Used by Zephyr at build time to generate hardware abstraction code.
- Files: *.dts (base), *.overlay (app-level override)



DeviceTree

Nodes

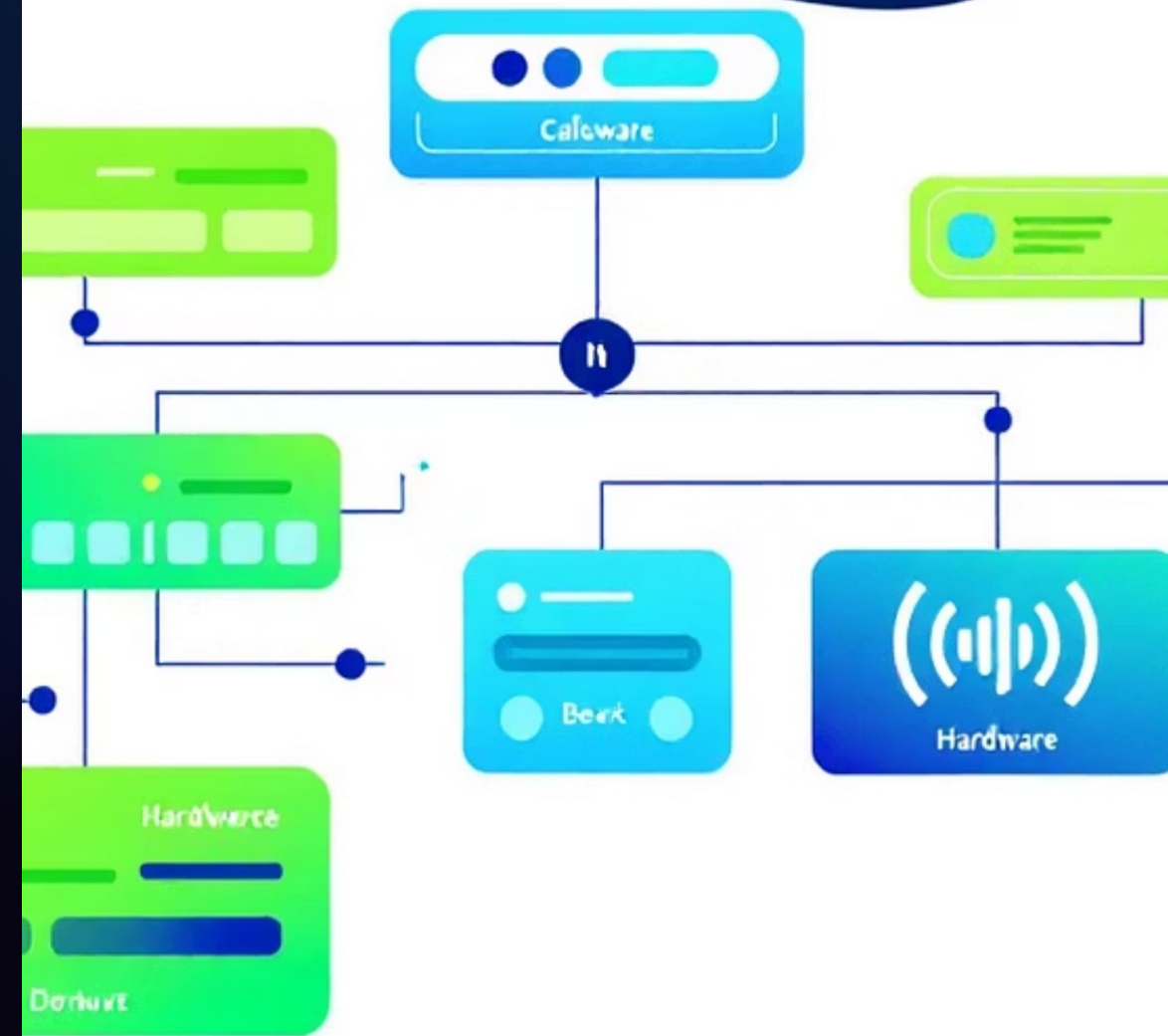
Hardware blocks (e.g.,
&gpio0)

Properties

Configuration data (pins,
levels)

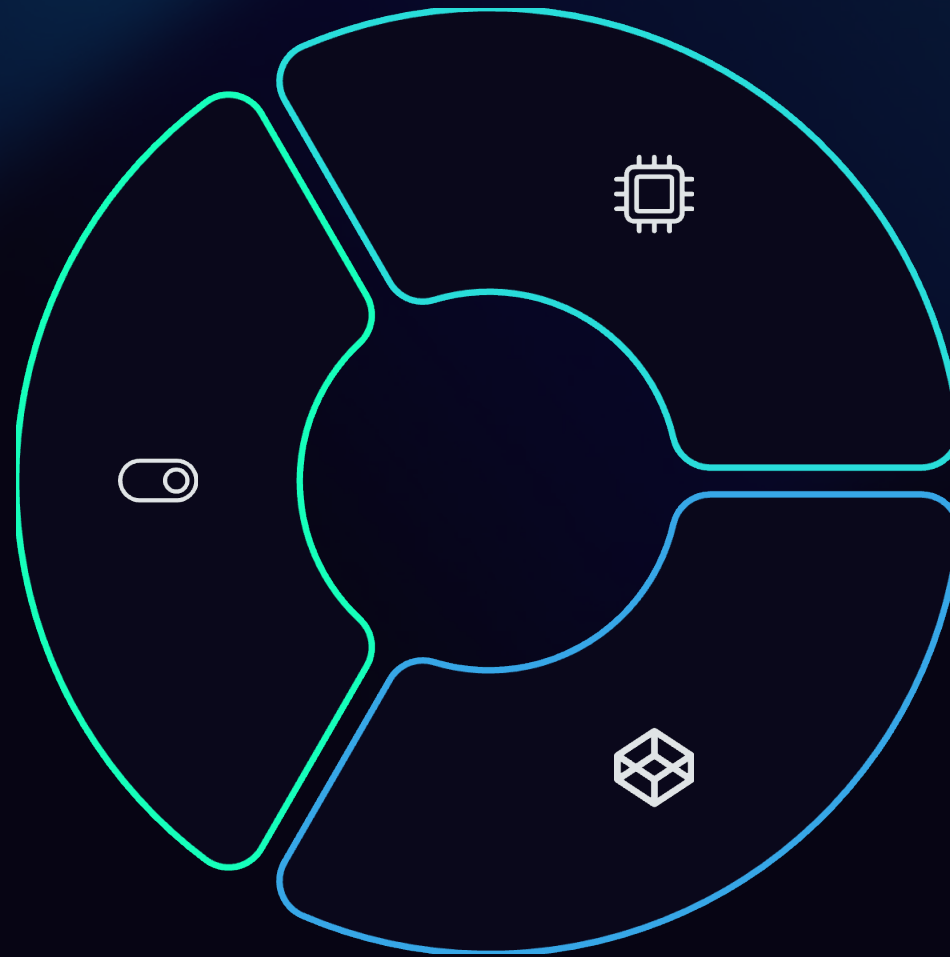
Labels & Aliases

Simplified code access



How Kconfig & DeviceTree Work Together

Kconfig
Enables drivers
(CONFIG_GPIO=y)



DeviceTree

Defines hardware (LED on
GPIO0, pin 13)

Application

Uses enabled, defined hardware



Live Example – LED

Blink

Start with Hello

World

Base project structure

Edit Files

prj.conf and main.c

Add app.overlay

Define LED alias

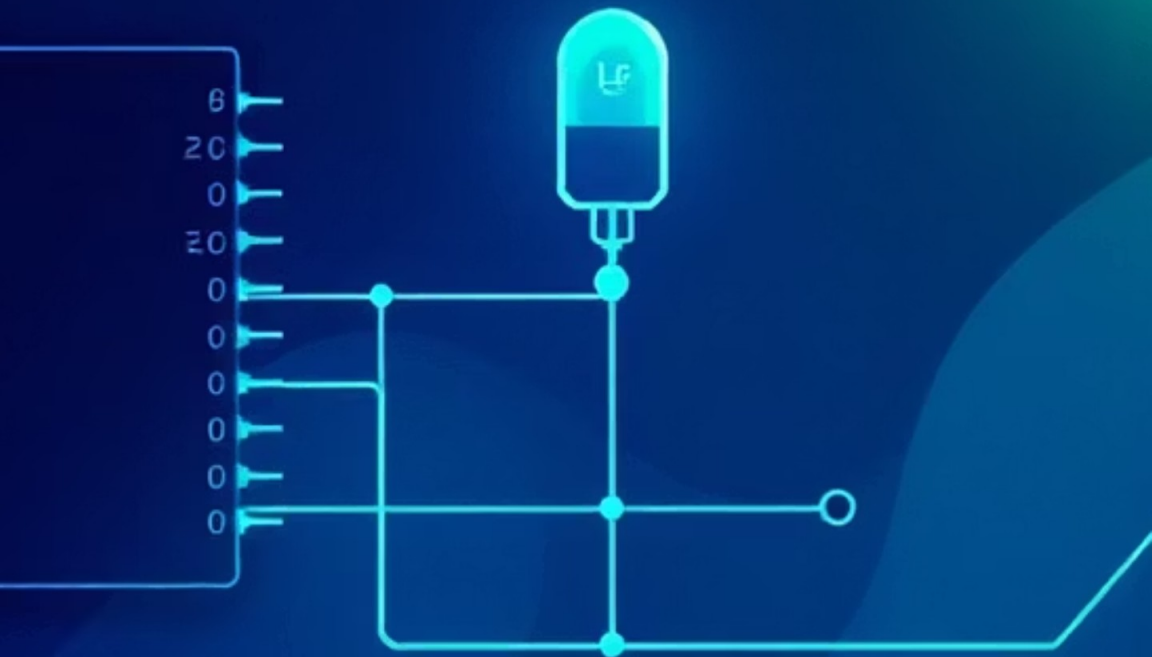
prj.conf – Example

```
CONFIG_GPIO=y
```

Enables the GPIO driver subsystem



```
1 emmeded systems
2
3 clunfntione fllle (;
4   configuration for suruan );
5   contlao settings
6   configured etting;
7
8 / emmeded system emftigvation );
9 emmeded system:
10 createlér {
11   rAnneCOWnd*-Cúúinn'rólers'();
12
13   onetlr:rol({})
14   ine ffastu, =deturauu #ll/);
15   emmeded set:file;
16   corader file rolerrll;
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18   conttater, celltton systom;
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20   grestatinn; );
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22   colected's entrtrigut on );
23   coteradamionc, eddelling);
24   recánee fceen;
25 }
26
27 frien kile codaer (ald emmeded systems; setting ); {
28 };
```

app.overlay -

```
/ {  
    aliases {  
        led0 = &led0;  
    };  
};
```

Defines hardware reference for LED

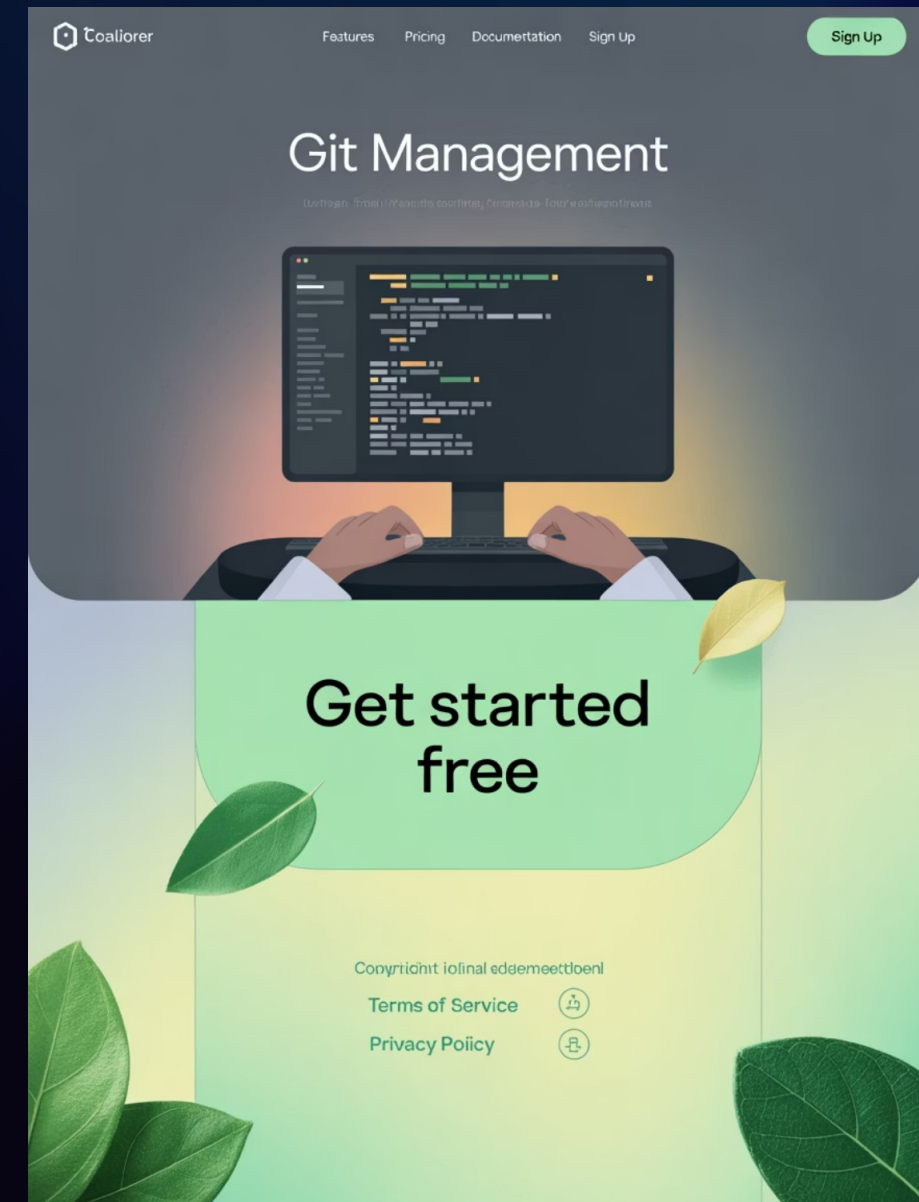
main.c – DeviceTree Usage

Add the code from blinky led sample to main.c



Run the sample

- build and flash on nrf52dk
files .dts and _defconfig from the folder
boards/nrf52dk generates the configuration
- Check the file build/zephyr/.config
- Check the file build/zephyr/zephyr.dts



Explore menuconfig

Run: `west build -t guiconfig`

Enable or disable options

Compare output with `build/zephyr/.config` now